

1. Project Goal and Business Questions

Cyclistic's Customer Growth Team is working on the next year business plan for New York city. The main task standing before them is growing Cyclistic Customer base. For this purpose they want to know:

- How does the demand on different bike stations look like?
- How do different types of riders use Cyclistic bikes?
- How weather and time conditions affect bike usage?

These questions define the structure of the analysis and form the requirements for the data we use in the project.

2. Data sources and initial assumptions

Primary dataset: NYC Citi Bike Trips

Secondary dataset: Census Bureau US Boundaries

The first dataset provides detailed information about bike trips in New York city (trip start date, trip end date, trip duration, trip starting and ending stations, user type and so on). The second table helps to conduct analysis on higher district level (zip code areas, neighborhoods and boroughs)

3. Pipeline and Target Tables Design Decisions

Our project is based on 3 target tables:

- Target table 1 contains data needed for trip growth year over year, weather condition and district demand analysis. The volume of data was big (17.5 million rows). That's why the initial target table was split into 3 parts and then united in Tableau interface
- Target table 2 focuses on congestion level research based on station location. Geographical data takes up a lot of space. Station longitude and latitude coordinates in addition to zip code data can make the single target table too enormous. This can slow down loading and visualization processes. So, a decision was made to create 2 separate tables: first for district based analysis (target table 1), second for station based research (target table 2)
- Target table 3 contains data for analysis daily usage dynamics on an hourly basis. Target tables 1 and 2 contain only the date component of each trip, without time. Because date without time simplifies aggregation and joining. So, we need to create the separate target table containing time of each trip start and end

4. Dashboard Design Choices

The final project includes a variety of visualizations:

- Maps with tables: maps increases geographical understanding of insights, tables provides additional information for easy access
- Line graphs for demonstrating the trip dynamics year over year
- Area chart showing bike usage time changes during the day
- Bar charts for understanding the impact of weather conditions on bike trip amount

5. Key insights and limitations

Key insights:

- The total trip amount is geographically unevenly distributed (more in Manhattan, less in Queens and Brooklyn)
- Weather conditions highly affect the number of trips (low temperature or any level of precipitation greatly reduce the total trip counts)
- The amount of trips increases noticeably in the morning and in the evening (clients actively use bikes for commuting)

Limitations:

- We can get more insights if have data about bike availability on each station at any time period
- Analyzing bike id and bike type can helps us to determine which type of bikes our clients prefer and which bikes require replacement or maintenance